

AfriSpine

Vercel Deployment Guide

Production deployment instructions for afri-spine.com
Including DNS configuration, database migration, and environment setup

Quick Reference	Key URLs
Framework: Next.js 16 + TypeScript Database: SQLite (local) / Turso libSQL (production) Hosting: Vercel Edge Network Domain: afri-spine.com	Dashboard: vercel.com/dashboard Deploy: vercel.com/new DNS: Your registrar panel Turso: turso.tech/dashboard

Table of Contents

- 1. Prerequisites** - Account setup and tools needed
- 2. Database Migration (SQLite to Turso)** - Setting up a production-ready cloud database
- 3. GitHub Repository Setup** - Push code and connect to Vercel
- 4. Vercel Project Configuration** - Create project and set environment variables
- 5. DNS Configuration** - Connect afri-spine.com with required records
- 6. Post-Deployment Verification** - Test all routes and functionality
- 7. Troubleshooting** - Common issues and solutions
- A. Environment Variables Reference** - Complete list of all configuration variables

1. Prerequisites

Before deploying AfriSpine to production, you need to set up the following accounts and tools. Each service has a free tier sufficient for initial deployment, with paid plans available as your user base grows. The total setup time is approximately 30-45 minutes if you already have accounts, or 60-90 minutes if starting from scratch.

1.1 Required Accounts

1 Vercel Account

Sign up at vercel.com using your GitHub account. The Hobby plan is free and supports custom domains, SSL certificates, and serverless functions with 100GB bandwidth per month. This is more than sufficient for launch.

2 GitHub Account

Create a repository at github.com/new. The codebase will be pushed here so Vercel can automatically build and deploy on every push. GitHub also serves as your code backup and collaboration platform.

3 Turso Database

Sign up at turso.tech and create a database. Turso provides libSQL, a production-ready fork of SQLite designed for serverless environments. The free tier includes 9GB storage and 500 databases, which is ideal for launching AfriSpine.

4 Paystack Account

Create a business account at paystack.com if you plan to process payments. You will need both the public and secret API keys for the production environment. Test keys work during development, but live keys are required for production transactions.

5 Domain Registrar Access

You need access to the DNS management panel for afri-spine.com. This is typically where you purchased the domain (Namecheap, GoDaddy, Cloudflare, etc.). You will add DNS records to point the domain to Vercel.

1.2 Local Tools

You need Git installed on your machine to push the codebase to GitHub. On macOS, Git comes with Xcode Command Line Tools. On Windows, download Git from git-scm.com. On Linux, install via your package manager (e.g., `sudo apt install git`). You also need Node.js 18+ and npm, which are required for the Vercel CLI if you prefer CLI deployment over the web dashboard.

2. Database Migration: SQLite to Turso

AfriSpine uses SQLite locally, but Vercel's serverless functions have an ephemeral filesystem, meaning SQLite data would be lost on every function invocation. Turso (libSQL) is the recommended production database because it is a drop-in replacement for SQLite that works over HTTP in serverless environments. The migration involves creating a Turso database, updating the Prisma schema, and seeding the production data.

Critical: Why Not SQLite on Vercel?

Vercel serverless functions use an ephemeral read-only filesystem. Each function invocation gets a fresh, isolated filesystem. SQLite writes to a local file, so any data written during one request will not be available in the next request. This makes SQLite fundamentally incompatible with Vercel's serverless architecture. Turso solves this by providing a remote libSQL database accessible over HTTP.

2.1 Create a Turso Database

Open your terminal and install the Turso CLI, then create a database and get your connection string. The following commands will set up everything you need. Replace "afri-spine-db" with your preferred database name. The location flag determines where your data is stored physically; choose a region close to your target users for the lowest latency.

```
$ curl -sSfL https://get.tur.so/install.sh | bash $ turso db create afri-spine-db --location iad $ turso db show afri-spine-db --url # Returns: libsql://afri-spine-db-yourname.turso.io $ turso db tokens create afri-spine-db # Returns: your-auth-token-here
```

2.2 Update Prisma Schema

Open prisma/schema.prisma and change the datasource provider from "sqlite" to "libsql" (Turso's provider name in Prisma). This is a single-line change that tells Prisma to use the Turso client instead of the local SQLite file. The rest of your schema remains identical because libSQL is fully compatible with SQLite.

```
datasource db { provider = "libsql" url = env("DATABASE_URL") }
```

2.3 Set DATABASE_URL

The DATABASE_URL for Turso follows a specific format that includes both the database URL and an authentication token as a query parameter. You will set this in Vercel's environment variables in the next section. The format is shown below, where you replace the placeholders with the values from Step 2.1.

```
DATABASE_URL="libsql://afri-spine-db-yourname.turso.io?authToken=your-turso-auth-token"
```

2.4 Run Migrations

After updating the Prisma schema and setting the DATABASE_URL, you need to push your schema to the Turso database and generate the Prisma client. Run these commands locally with the production DATABASE_URL set in your environment. This will create all the tables (Sender, Transaction, DigestIssue, DigestStory, ChamaCircle, and all other models) in your Turso database.

```
$ export DATABASE_URL="libsql://afri-spine-db-yourname.turso.io?authToken=..." $ npx prisma db push $ npx prisma generate
```

2.5 Seed Production Data (Optional)

If you want to populate the production database with sample data (Digest issues, stories, subscriptions), you can run the seed script. This is useful for testing the full application before going live. The seed script creates 4 Digest issues, 21 stories, 4 sponsors, 15 subscriptions, and 2 advertisers.

```
$ npx tsx scripts/seed-digest.ts
```

3. GitHub Repository Setup

Pushing your codebase to GitHub enables Vercel's automatic deployment pipeline. Every push to the main branch triggers a new build and deployment. This section covers creating the repository, pushing the code, and connecting it to Vercel. If you already have the codebase locally as a ZIP file, extract it first before proceeding with these steps.

3.1 Create a New GitHub Repository

Navigate to github.com/new and create a new repository. Name it "afrispine" or "afri-spine". Keep it Private to protect your source code. Do not initialize with a README, .gitignore, or license, since the codebase already includes these files. After creation, GitHub will show you the remote URL to push to.

3.2 Push the Codebase

If you have extracted the ZIP file locally, navigate to the project directory and run the following commands. These initialize a Git repository (if not already), add all files, commit, and push to GitHub. The codebase is already cleaned and ready for production, with a proper .gitignore that excludes build artifacts, database files, and development tools.

```
$ cd afrispine $ git init $ git add -A $ git commit -m "Initial commit: AfriSpine production codebase" $ git remote add origin https://github.com/YOUR_USERNAME/afrispine.git $ git branch -M main $ git push -u origin main
```

3.3 Alternative: GitHub CLI

If you have the GitHub CLI (gh) installed, you can create the repository and push in a single command. This is faster and avoids switching between the browser and terminal. The gh CLI also handles authentication automatically if you have previously logged in.

```
$ gh repo create afrispine --private --source=. --push
```

4. Vercel Project Configuration

With your code on GitHub and database on Turso, the next step is to create a Vercel project and configure it. Vercel will automatically detect the Next.js framework, install dependencies, and build the project. You need to set environment variables and configure the build command to include Prisma generation.

4.1 Import from GitHub

Navigate to vercel.com/new and click "Import Git Repository". Select the afrispine repository you just created. Vercel will automatically detect Next.js as the framework and configure the build settings accordingly. The default build command (next build) and output directory (.next) are correct, but you need to modify the build command to include Prisma generation.

4.2 Configure Build Settings

In the "Build and Output Settings" section, update the Build Command to include Prisma generation. This ensures that the Prisma client is generated before the Next.js build starts, which is required for all database operations. The install command should remain as the default (npm install for npm, or pnpm install if using pnpm).

```
# Build Command: npx prisma generate && next build # Install Command (default): npm install
# Output Directory (default): .next
```

Build Command Note

The vercel.json file included in the codebase already specifies the build command as "npx prisma generate && next build". Vercel will read this configuration automatically. However, you can also set it in the Vercel dashboard under Settings > General > Build Command if you prefer to manage it there.

4.3 Set Environment Variables

Navigate to Settings > Environment Variables in your Vercel project dashboard. Add each variable from the table below. For the DATABASE_URL, use the Turso connection string from Section 2.1. For JWT_SECRET and NEXTAUTH_SECRET, generate strong random strings using the command shown below. These secrets are critical for security, so use cryptographically random values.

```
# Generate random secrets: $ node -e
"console.log(require('crypto').randomBytes(32).toString('hex'))"
```

Variable	Required	Description
DATABASE_URL	YES	libSQL connection string (Turso) or PostgreSQL URL
JWT_SECRET	YES	32+ character random string for JWT tokens
NEXTAUTH_SECRET	YES	32+ character random string for NextAuth sessions
NEXTAUTH_URL	YES	https://afri-spine.com
PAYSTACK_SECRET_KEY	Recommended	Paystack live secret key for payments
PAYSTACK_PUBLIC_KEY	Recommended	Paystack live public key

Variable	Required	Description
FLUTTERWAVE_SECRET_KEY	Optional	Flutterwave secret key
NEXT_PUBLIC_APP_URL	YES	https://afri-spine.com
NEXT_PUBLIC_APP_NAME	YES	AfriSpine
ADMIN_EMAIL	Recommended	admin@afri-spine.com

5. DNS Configuration for afri-spine.com

Connecting your custom domain to Vercel requires adding specific DNS records at your domain registrar. These records tell the global DNS system that afri-spine.com should resolve to Vercel's servers. Vercel then routes the request to your specific project. This section provides the exact records to add and the order in which to add them. The DNS propagation typically takes 5-30 minutes, though in rare cases it can take up to 48 hours.

5.1 Add Domain in Vercel

First, add your domain in the Vercel dashboard. Navigate to your project Settings > Domains and click "Add Domain". Enter "afri-spine.com" as the primary domain. Vercel will display a list of required DNS records. These records must match exactly what is shown in Section 5.2 below. After adding the records in your DNS panel, click "Verify" in Vercel to confirm the domain connection.

5.2 Required DNS Records

Add the following DNS records at your domain registrar (where you purchased afri-spine.com). The exact interface varies by registrar, but the values are the same everywhere. If your registrar does not support the TXT record for SPF, you can skip it initially and add it later when setting up email.

Type	Name	Value	TTL
A	@	76.76.21.21	300
CNAME	www	cname.vercel-dns.com	300
CNAME	_vercel	cname.vercel-dns.com	300
TXT	@	v=spf1 include:_spf.google.com ~all	3600
MX	@	10 mail.protonmail.ch	3600
MX	@	20 mailsec.protonmail.ch	3600

Important Notes on DNS

1. The A record value (76.76.21.21) is Vercel's load balancer IP. Do not use a different IP.
2. If your registrar requires a trailing dot on CNAME values (e.g., `cname.vercel-dns.com.`), add it.
3. The `_vercel` CNAME record is required for domain verification and SSL certificate provisioning.
4. The MX and TXT records are for email (ProtonMail example). Adjust if using a different provider.
5. DNS propagation can take up to 48 hours, but typically completes within 15 minutes.
6. Do NOT add these records until Vercel has created your project deployment.

5.3 SSL Certificate

Vercel automatically provisions a free SSL certificate from Let's Encrypt once your DNS records are correctly configured and verified. This happens automatically within minutes of DNS propagation completing. You do not need to manually configure SSL. Vercel also handles certificate renewal automatically. The certificate covers both afri-spine.com and www.afri-spine.com, ensuring all traffic is encrypted.

5.4 DNS Registrar-Specific Instructions

Below are quick links to DNS management for common registrars. Navigate to the DNS management page and add the records from Section 5.2. Each registrar has a slightly different interface, but the record types, names, and values are identical regardless of your provider.

Registrar	DNS Path	Notes
Cloudflare	DNS > Records	No proxy (grey cloud)
Namecheap	Domain List > Manage > Advanced DNS	Add new records manually
GoDaddy	My Products > DNS	Add records in the DNS management zone file
Route 53	Hosted zones > afri-spine.com	Create records in the zone
Google Domains	DNS > Custom records	Add each record type individually

6. Post-Deployment Verification

After Vercel completes the deployment and DNS is configured, you need to verify that everything works correctly. This section provides a checklist of all routes and features to test. Open a browser and navigate to <https://afri-spine.com> to begin testing. The verification process takes approximately 10-15 minutes and ensures a smooth user experience at launch.

6.1 Core Route Verification

Test each of the following routes to ensure they return a 200 OK response and render correctly. Pay special attention to the Digest pages, which use a magazine-style layout with cream backgrounds and gold accents. The admin panel requires authentication, so verify the login flow separately.

Route	Expected Page	Verification
/	Homepage with SPA routing	All navigation links work
/digest	Digest current issue	Magazine layout renders with stories
/digest/archive	Digest archive	Past issues display correctly
/digest/subscribe	Subscription page	Form submits to Paystack
/digest/advertise	Advertise with us	Ad tiers display, booking flow works
/admin/digest	Digest admin panel	Login required, 5 tabs visible
/send	Send money page	Form renders, quote API responds
/wealth	Wealth dashboard	Portfolio and market data load
/api/digest/issues	Issues API	Returns JSON with issue data

6.2 Database Connectivity

After deployment, verify that the Turso database is correctly connected by visiting the API endpoints that query the database. The `/api/digest/issues` endpoint should return JSON data from your database. If it returns a 500 error, check that the `DATABASE_URL` environment variable is correctly set in Vercel and that the Prisma schema has been pushed to Turso.

```
# Test database connectivity: curl https://afri-spine.com/api/digest/issues # Should return JSON: {"issues": [...]}
```

6.3 SSL and Security

Verify that HTTPS is working correctly by visiting <https://afri-spine.com>. The browser should show a padlock icon indicating a valid SSL certificate. Also check that HTTP requests are automatically redirected to HTTPS. Vercel handles this automatically, but it is worth verifying. Additionally, check the security headers configured in `vercel.json` by inspecting the response headers in your browser's developer tools.

7. Troubleshooting

This section covers the most common issues encountered during deployment and their solutions. If you encounter an issue not listed here, check the Vercel deployment logs in your project dashboard under Deployments > [latest] > Build Logs and Function Logs.

Issue	Solution
Build fails with "prisma generate" error	The Prisma schema has a syntax error or the provider is not recognized. Ensure the provider is set to "postgresql".
Pages return 500 on database queries	The DATABASE_URL is either not set in Vercel environment variables, or the Turso auth token is invalid.
Custom domain shows "Domain Not Found"	DNS records have not propagated yet. Wait 15-30 minutes and try again. Use "dig afri-spine.com" or "nslookup afri-spine.com".
SSL certificate not provisioning	Vercel provisions SSL automatically after DNS verification completes. Ensure all required DNS records are correctly set.
Fonts or images not loading	Check that the /public directory is committed to Git and included in the deployment. Vercel serves static files from this directory.
API routes return 404	Ensure the vercel.json does not have conflicting rewrites. The existing rewrite rule catches all non-API requests.
Build exceeds memory limit	The project has been optimized to build within Vercel's default memory limits. If you add heavy dependencies, consider using a larger build plan.

Appendix A: Environment Variables Reference

This is the complete list of all environment variables used by AfriSpine. Only the variables marked as "Required" are needed for the initial deployment. Additional variables can be added later as you enable specific features like payments, KYC, WhatsApp notifications, and email delivery.

Variable	Category	Description	Example
DATABASE_URL	Required	libSQL/PostgreSQL connection string	libsql://...
JWT_SECRET	Required	JWT token signing secret (32+ chars)	crypto random
NEXTAUTH_SECRET	Required	NextAuth session encryption key	crypto random
NEXTAUTH_URL	Required	Canonical URL of your app	https://afri-spine.com
NEXT_PUBLIC_APP_URL	Required	Public-facing app URL	https://afri-spine.com
NEXT_PUBLIC_APP_NAME	Required	Application display name	AfriSpine
PAYSTACK_SECRET_KEY	Payments	Paystack live secret key	sk_live_...
PAYSTACK_PUBLIC_KEY	Payments	Paystack live public key	pk_live_...
FLUTTERWAVE_SECRET_KEY	Payments	Flutterwave secret key	FLWSECK-...
FLUTTERWAVE_PUBLIC_KEY	Payments	Flutterwave public key	FLWPUBK-...
SMILE_ID_PARTNER_ID	KYC	Smile Identity partner ID	From dashboard
SMILE_ID_API_KEY	KYC	Smile Identity API key	From dashboard
WHATSAPP_PHONE_NUMBER_ID	Messaging	WhatsApp Business phone ID	From Meta
WHATSAPP_ACCESS_TOKEN	Messaging	WhatsApp access token	From Meta
WHATSAPP_VERIFY_TOKEN	Messaging	Webhook verification token	Custom string
RESEND_API_KEY	Email	Resend email API key	re_...
ADMIN_EMAIL	Admin	Admin user email address	admin@afri-spine.com
ADMIN_PASSWORD_HASH	Admin	Bcrypt hash of admin password	\$2b\$10\$...

Generated for AfriSpine Engineering Team. For support, refer to the Vercel documentation at vercel.com/docs.